PRODUCT REQUIREMENTS DOCUMENT · V3.0 · STATUS & SUBMISSION GAP ANALYSIS

MPSTME GUARDIAN

What is built, what is left, and everything still owed for the course faculty's MDRIIA Capstone submission

MuJoCo · Dual-UAV Surveillance · MDRIIA Group 05

> *"To what extent can a collaborative dual-UAV surveillance system simulated in MuJoCo optimize campus perimeter patrol cycle time and OpenCV human detection latency compared to static security guard patrols?"*

COURSE	MDRIIA — Modern Day Robotics & Its Industrial Applications (702CO0E012) · Capstone ICA, 20 marks
FACULTY	The course faculty, SVKM's NMIMS MPSTME, Mumbai
GROUP	05 — E043 Muaaz Shaikh (Lead, owner of `main`) · E051 Aditya Rajkumar · E075 Zaid Rezaur Rahman · E077 Soumya Subhankar Ranasingh
STATUS AS OF	Friday 30 October 2026 (v3.1; v3.0 was Thursday 29 October) — **Dossier deadline (Mon 26 Oct, 23:59 IST) has passed · Expo & Viva: Saturday 31 October 2026**
REQUIREMENT SOURCES	Grading Rubric & Evaluation Policy · LinkedIn & GitHub Showcase Guide · Group 05 Technical Dossier template · Academic & Industry Standards Manual · Contributing / Branching guides · Group 05 scaffold README (course monorepo, read 29 Oct 2026)
SUPERSEDES	PRD v3.0 status (29 Oct) and PRD v2.0 (8 Oct 2026) for status and submission scope; v2.0 remains the design reference. Faculty rulings of 29 Oct are in §15 and override earlier text

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LEGEND. DONE implemented and checked · PARTIAL exists but does not yet meet the requirement · NOT STARTED nothing usable yet · DECISION needs a ruling (team or faculty) · DRAFTED drafted for a teammate or in the fill sheet; they must apply, commit and sign · HUMAN only a person can do it. Priorities: **P0** = marks are lost or the submission is invalid without it · **P1** = needed for full marks in a criterion · **P2** = quality / viva safety.

1 Executive Summary

The engineering core — a MuJoCo model of the NMIMS Main Building, two flying quadrotors with a cascaded controller, collision-safe coordination and a provenance-gated analytics pipeline — is built and working in both scenes (interior and exterior, all patrolling inside the college), and the 500 Hz real-trial dataset (180 runs, N = 30 in each of six cells, timestep 0.002 s) is committed with figures regenerated from it. The three teammates' code regions, statistics and business model exist as drafted patches that each member must apply and commit from their own account. Still owed by people: Word dossier fill and sign-off, the LinkedIn video, peer review, SAP IDs, and the Teams turn-in. The dossier and video deadlines have already passed.

1 day
TO EXPO & VIVA (SAT 31 OCT)

4 days
DOSSIER OVERDUE (DUE 26 OCT)

1 / 4
REGIONS COMMITTED (E043); 3 DRAFTED AS PATCHES, ALL GATES OK

180
REAL TRIALS IN `*_REAL_TRIALS.CSV` (6 CELLS X 30)

0 / 6
SECTIONS FILLED IN WORD (AUTOFILL DRAFT + FILL SHEET READY)

The five things that decide the marks

#	ITEM	MARKS IT FEEDS	STATUS	OWNER
1	Filled Technical Dossier (.docx + .pdf) committed to `docs/` and turned in on MS Teams by **each** member	All 20 (no turn-in ⇒ no marks for that student)	DRAFTED, HUMAN FILL	All · E043 compiles
2	N ≥ 30 (manual: ≥ 50) seeded MuJoCo runs per condition, 500 Hz, in `analytics/campus_patrol_real_trials.csv`, with Welch t-test, Cohen's d, 95 % CI	C1 Telemetry 2 + Statistics 2	DATA DONE · STATS: PATCH	E043 + E075
3	CSBS script computing H_reclaimed, P_payback, NCRI/κ, M/M/c delay, FMEA, sensitivity graph — no hard-coded results, no monetary values	C2 — all 5	DRAFTED (PATCH)	E077
4	Surveillance FSM (PATROL → DETECT → VERIFY → ALERT → TRACK → RETURN) with E051 detector and E075 geofence implemented	C1 Control 2 + viva for E051/E075	FSM DONE; E051/E075 IN PATCHES	E051, E075, E043
5	3–4 min LinkedIn video (live sim, business graph, all four on webcam), AI audit ×3, peer review of Groups 06 & 11	C3 — 2 of 5	HUMAN	All

IMMEDIATE ACTION: Q1 (late dossier) is still unanswered; send the drafted message in `docs/submission/email_to_sir_DRAFT.md` yourself. Then each member applies their handoff patch, commits from their own account, fills their Word cells and records their video part (§6, status column).

2 What Sir Requires — the Submission Contract

Consolidated from the Grading Rubric, the Showcase Guide and the Dossier template. These are the deliverables that are *checked*; anything not listed here earns no marks on its own.

#	DELIVERABLE	WHERE IT GOES	DEADLINE	STATUS
D1	Technical Dossier completed in **Microsoft Word desktop** (drop-downs and "type here" fields kept intact, not Google Docs), exported to PDF	`Group_05_…/docs/MDRIIA_Group_05_Technical_Dossier.docx` **and** `.pdf`, original file name	Mon 26 Oct 23:59 — **passed**	DRAFT + FILL SHEET READY; WORD FILL IS HUMAN
D2	Dossier PDF attached and **"Turn in" clicked individually** by all four members	MS Teams assignment	Mon 26 Oct — **passed**	HUMAN
D3	Group folder in the **course monorepo**: MJCF model, controller, business script, `*_real_trials.csv`	`github.com/sunny-nanade/NMIMS-MPSTME-MDRIIA-2026` via PR into `main`	With D1	PARTIAL — export prepared (`docs/submission/monorepo_pr_steps.md`), PR not opened; work is only in the private `MuaazSM/mdriia-project`
D4	3:00–4:00 LinkedIn video, uploaded natively, one post per group, teammates + MPSTME + the course faculty tagged, monorepo folder linked	LinkedIn; URL in dossier §5 and `docs/TEAM_ROSTER.json → linkedin_url`	Sun 25 Oct 18:00 — **passed**	HUMAN (SCRIPT DRAFTED)
D5	Peer review of Group 06 and Group 11 videos (0–2 on Problem, Physics, Business, Data, Clarity + best moment + one viva question)	Dossier §5.2	With D1	HUMAN (TEMPLATE DRAFTED)
D6	Live Expo demo: simulation running from a **fresh `git clone` + `pip install -r requirements.txt`**, FSM states visible, one handled failure shown; business graphs; webcam PiP of every member; **no PPT**	Studio Showcase	Sat 31 Oct	PARTIAL
D7	Individual viva: own role and code *and* the whole project including the business model	Viva voce	Sat 31 Oct	PARTIAL

Rules that can zero a component

RULE	WHAT IT MEANS FOR US	STATUS
Data provenance — `SIMULATED_*` and starter figures are `np.random` targets	The starter data was renamed and the figure script reads only the real-trials file, which carries a provenance JSON.	DONE
No hard-coded results in the business script	E077's patch rewrites `campus_security_economics.py`; the audit found none of the old constants in the patched script. The E043 branch still holds the old file until the patch is applied.	DRAFTED (PATCH)
Zero-Money Rule — ratios only (κ, NCRI, P_payback)	The integrity audit found no monetary values in any deliverable after cleanup.	DONE (AUDIT)
Every claim/citation verified at its primary source (§1.1)	Claims table drafted in the fill sheet with sources from `docs/research/sources_verified.md` and E077's CLAIMS_AND_BASELINE.md; the "45 to 8.5 min" figure is not used anywhere.	DRAFTED

RULE	WHAT IT MEANS FOR US	STATUS
AI use logged (§5.1); undisclosed AI content = integrity breach	Each packet has an AI_DISCLOSURE.md and the fill sheet has the §5.1 cases; every member must sign their own entry.	HUMAN
Attribution matches git history (checked with `git shortlog -sn -- Group_05_*/`)	E043 commits all come from the noreply author. Each teammate must apply their patch and commit from their own account (COMMITS.md).	HUMAN

3 Marks Map: 20 Marks, Criterion by Criterion

Component 1 — CDIO Robotics Engineering & MuJoCo Physics (10)

CRITERION	MK	REQUIRED EVIDENCE	WHERE WE ARE	STATUS
Problem framing	2	Authorised title; verified baseline friction; governing standards with clauses; claim verification table (§1)	Title fixed; sourced guard-baseline inputs, claims table and standards decisions drafted in the fill sheet.	DRAFTED
MuJoCo rigid-body model	2	`models/*.xml` loads; friction / `solref` / timestep justified; mechanism matches title	Loads in both scenes at 0.002 s; friction and solref justified in `docs/research/mjcf_parameter_justification.md`.	DONE
Closed-loop control	2	Controller with implemented FSM and stated control law; honest authorship table	Cascaded control law plus the surveillance FSM (state on every row); the detector and geofence come from the E051 and E075 patches.	DONE (WITH PATCHES)
Telemetry & Monte Carlo	2	500 Hz logging; N ≥ 30 randomised runs per condition, seeds recorded; tracking RMSE; control effort $\int \tau^2 dt$	500 Hz telemetry, tracking RMSE and control effort per run; 180 runs, N = 30 in each of 6 cells.	DONE
Statistical validation	2	Baseline vs proposed; t-test; Cohen's d; 95 % CI; plots regenerate from `*_real_trials.csv`	Statistics script drafted in E075's patch; figures regenerated from the real-trials CSV.	DRAFTED (PATCH)

Component 2 — CSBS Techno-Economic Business Systems (5) — owner E077

CRITERION	MK	REQUIRED EVIDENCE	WHERE WE ARE	STATUS
Labour reallocation & payback	1.5	H_reclaimed and dimensionless P_payback computed from sourced inputs	Drafted in E077's patch, computed from sourced inputs; tests pass.	DRAFTED (PATCH)
Cost reduction	1	NCRI = 1 − κ, dimensionless	NCRI and kappa drafted in E077's patch.	DRAFTED (PATCH)
Queuing model	1	M/M/1 or M/M/c with λ, μ, c, ρ stated	M/M/c with lambda, mu, c and rho drafted in E077's patch.	DRAFTED (PATCH)
Risk	1	FMEA ≥ 4 modes, RPN pre/post, mitigation shown in sim or code	Seven FMEA modes drafted in E077's patch; occurrence is "not measured" for two modes.	DRAFTED (PATCH)
Sensitivity	0.5	Graph on most uncertain input with break-even value	Sensitivity graph and queue graph drafted in E077's patch (300 DPI PNGs).	DRAFTED (PATCH)

Component 3 — Studio Showcase & Viva Voce (5)

CRITERION	MK	REQUIRED EVIDENCE	WHERE WE ARE	STATUS
LinkedIn video (3–4 min)	1	Live sim, business graph, every member on webcam PiP	Script and shot list drafted; nothing recorded yet.	HUMAN
AI audit & peer review	1	Three verified AI-error cases; review of Groups 06 and 11	Three AI cases drafted in the fill sheet, more in `ai_audit_candidates.md`; peer-review template drafted; reviews not done.	HUMAN
Live Expo demo	1	Runs from fresh clone; FSM states shown; one handled failure	Fresh clone passed; viewer overlay shows each UAV's FSM state; head-on demo ready. The overlay drawing was not checked by screenshot.	DONE
Individual viva	2	Each member defends own role, code, and the whole project	Viva prep files written for all four roll numbers; each member must rehearse.	HUMAN

4 Product Requirements & Status

Functional (FR) and non-functional (NFR) requirements of the product as it must exist on Saturday. Each row names the acceptance test that closes it.

ID	REQUIREMENT	ACCEPTANCE TEST	OWNER	STATUS
FR-01	MJCF model of the NMIMS Main Building (interior default, exterior variant) with two quadrotors	`mujoco.MjModel.from_xml_path(...)` loads; `src/test_env.py` passes	**E043**	DONE
FR-02	MJCF timestep 0.002 s (500 Hz) with friction and solref justified	Option reads 0.002; hover, patrol and separation test still pass; justification text in dossier §2.1	**E043**	DONE
FR-03	Cascaded closed-loop flight control with stated gains (units)	Gains table in dossier §2.1; altitude hold within ±0.01 m on the ground loop	**E043**	DONE
FR-04	Coordinated dual-UAV patrol (sectors / offset ring) covering the corridor loop and atrium galleries	Both policies run; revisit and coverage metrics reported	**E043**	DONE
FR-05	Inter-UAV collision avoidance, $d_{min} \geq 2.5$ m	`tests/test_e043_separation.py` : layered method passes head-on	**E043**	DONE
FR-06	OpenCV HSV intruder detection on the onboard camera, bounding box + latency	`detect_intruder_hsv` returns bbox; IoU ≥ 0.70 vs segmentation ground truth at noon; latency logged	**E051**	DRAFTED (PATCH)
FR-07	Inverse pinhole projection pixel → world (f = 450 px, 640 × 480)	Projected intruder position within stated error of true position	**E051**	DRAFTED (PATCH)
FR-08	Restricted-zone geofence (ray-casting point-in-polygon), 4-tier alert with hysteresis, GPS/IMU noise injection, breach log	`evaluate_geofence_status` returns tiers 0–3 for test points; breach logged with timestamp	**E075**	DRAFTED (PATCH)
FR-09	Surveillance FSM: PATROL → DETECT → VERIFY → ALERT(0–3) → TRACK → RETURN_TO_PATROL	State logged every row; state shown as overlay in the live viewer; one transition demonstrable live	**E043 integrates**	DONE (WITH PATCHES)
FR-10	500 Hz telemetry: t, run_id, seed, pose, velocity, reference, tracking error, actuator forces, FSM state, events	One CSV per run; row interval 0.002 s	**E043 / E075 QA**	DONE
FR-11	Per-run tracking RMSE and control effort $\int \tau^2 \, dt$	Columns present in `campus_patrol_real_trials.csv`	**E043**	DONE
FR-12	Monte Carlo runner: N runs per condition, randomised wind / intruder timing / start, seeds recorded	`analytics/campus_patrol_real_trials.csv` + provenance JSON	**E043**	DONE
FR-13	Statistics: Welch t-test, Cohen's d, 95 % CI, baseline vs proposed	`analytics/statistical_validation.py` prints t, df, p, d, CI from the CSV	**E075 / E043**	DRAFTED (PATCH)
FR-14	Publication figures regenerate from real-trials CSV only	`generate_paper_figures.py` produces 3 figures with no watermark	**E043**	DONE
FR-15	CSBS model: H_reclaimed, P_payback, T_payback, NCRI/κ, M/M/c, sensitivity ±30 %, break-even	Script re-run reproduces every number in dossier §3.1 from sourced inputs	**E077**	DRAFTED (PATCH)

ID	REQUIREMENT	ACCEPTANCE TEST	OWNER	STATUS
FR-16	FMEA ≥ 4 failure modes with RPN before/after mitigation	Each mitigation points to code or a simulation run	**E077 + all**	DRAFTED (PATCH)
FR-17	Live viewer with FSM state and one handled failure	`mjpython src/view_live.py` shows states; head-on conflict resolved on screen	**E043**	DONE (OVERLAY NOT SCREENSHOT-CHECKED)
FR-18	Obstacle sensing and avoidance in the interior (and exterior) scene: every drone senses all obstacles, and avoidance is never disabled in any FSM state	`tests/test_e043_obstacle_perception.py` ; `tests/test_e043_surveillance_fsm.py` checks avoidance runs every step while tracking	**E043**	DONE
NFR-01	Runs from a fresh clone with `pip install -r requirements.txt`	Clean-directory clone on a second laptop (Windows or macOS)	**E043**	DONE (FRESH CLONE PASSED)
NFR-02	Data provenance: no `SIMULATED_*` or `np.random` number reported as a result	Figure gate refuses unprovenanced CSV; dossier provenance boxes ticked honestly	**All**	DONE (gate)
NFR-03	No monetary values anywhere (code, dossier, video)	a case-insensitive grep of the deliverables for monetary words, codes and symbols returns nothing	**E077**	DONE (AUDIT PASSED)
NFR-04	REP-103 frames and SI units (m, rad, m/s, N·m)	Stated in dossier §2	**E043**	DONE (MuJoCo ENU, SI)
NFR-05	Each member ≥ 4 conventional feature commits from own account on own branch	`git shortlog -sn -- Group_05_*/`	**All**	E043 DONE; 3 MEMBERS HUMAN
NFR-06	No videos, venvs or large binaries in the monorepo	MP4s stay out of the PR; video lives on LinkedIn	**E043**	DONE (EXPORT EXCLUDES MP4S)

5 What Is Done (Evidence)

AREA	DELIVERED	EVIDENCE IN THE REPO
Repository	Course scaffold layout preserved; pinned `requirements.txt`; README onboarding; private GitHub repo with `main` owned by E043	`README.md`, `requirements.txt`, commit `c3fdea7`
Scene	NMIMS Main Building interior: 10 floors × 3.6 m, central atrium 21.6 × 38.5 m (satellite-measured), 132 m ground-floor corridor loop, galleries, VIP Lounge (restricted zone), cafeteria, lift lobby, Wall of Fame; exterior campus variant from OpenStreetMap	`models/nmims_main_interior.xml`, `models/scene_source/*`, `interior_manifest.csv` (MEASURED vs ASSUMED)
Vehicles	Two 1.35 kg "+" quadrotors, arm 0.26 m, four rotors 0–7.5 N each with yaw drag torque, nadir and forward cameras (640 × 480, f = 450 px), IMU sensors	`models/uav_fleet.xml`
Control law	Position P → velocity PI → flat-output inversion → attitude PD with gyroscopic feed-forward; exact inverse mixer; tilt slew limiter; principal-frame inertia	`src/aerial_patrol_swarm.py` (E043 region)
Mission	Phases takeoff → transit → descend → patrol; policies `sectors` and `offset_ring`; per-UAV yaw modes	same
Safety	Altitude layers + closing-speed governor + right-hand sidestep on top of the guide's APF	`tests/test_e043_separation.py`: APF alone 0.44 m (contact) → layered 3.31 m (pass), head-on segment moved inside the compound
Disturbance	Ornstein–Uhlenbeck gust field written to MuJoCo wind	`_gust_step`
Metrics	Revisit intervals, gallery coverage, energy (Wh), collisions, altitude error	console summary + telemetry CSV
Analytics	Seeded batch runner with provenance (seeds, code/model hashes, versions); figure generator that refuses unprovenanced data	`analytics/run_benchmark.py`, `generate_paper_figures.py`
Team tooling	Region-ownership PR gate; live viewer; stills + 2 × 2 patrol video renderer	`src/check_region_ownership.py`, `view_live.py`, `render_scene_media.py`
Planning	PRD v2.0 and Implementation Plan v2.0 (exterior scene) with Mermaid sources	`docs/planning/`

> **DEVELOPMENT CHECKS, NOT RESULTS.** Single 240 s interior runs (one seed) gave: `sectors` worst ground-loop revisit 53.4 s, 66.7 % gallery coverage, 0 collisions; `offset_ring` 36.7 s and 0 % gallery coverage. These were development checks from before the Monte Carlo batch. They are **NOT** Monte Carlo evidence and must not appear in the dossier as results; dossier numbers come from the 180-run CSV only.

Work on other branches (reviewed, not merged)

E077's `feat/e077-guardian-live-surveillance-upgrade` (10 commits, 381 files, ≈ 86 k lines) adds a YOLO/ByteTrack runtime, A* navigation, a control-room page and an economics module in monetary units under `integrations/`. It is carefully caveated but: fails the ownership gate on all files, leaves `campus_security_economics.py` untouched, uses monetary units, is Windows/CUDA-tested only, and 8 of 10 commits carry a mistyped e-mail (`@mgail.com`) so GitHub does not attribute them to her account. Recommendation: park it; salvage only the "every assumption needs a provenance string" idea.

6 What Is Left — Master Checklist (Prioritised)

P0 — submission is invalid or a component scores zero without it

ID	TASK	OWNER	DONE WHEN	STATUS 30 OCT
L-01	Tell the course faculty the dossier is late; agree a turn-in time; raise §15 questions	E043	Written reply received	HUMAN: EMAIL DRAFTED, NOT SENT; Q1 OPEN
L-02	Copy the official `MDRIIA_Group_05_Technical_Dossier.docx`, `docs/README.md`, `docs/figures/README.md` from the monorepo into our `docs/`	E043	Files present, unmodified	DONE
L-03	Rename scaffold data to `analytics/SIMULATED_campus_patrol_benchmark.csv`; point runner output to `analytics/campus_patrol_real_trials.csv`	E043	Figure script reads only the real-trials file	DONE
L-04	Rewrite `campus_security_economics.py`: delete every hard-coded result; compute from sourced inputs and the real-trials CSV; ratios only	E077	No literal outputs; `grep` finds no monetary words	DRAFTED (E077 PATCH)
L-05	Run the Monte Carlo batch (baseline + proposed, N ≥ 30 each, seeds recorded)	E043	CSV + provenance JSON committed	DONE (180 RUNS)
L-06	Fill all six dossier sections and sign-off in Word desktop; export PDF; commit both	All · E043 compiles	.docx + .pdf in `docs/`	HUMAN (AUTOFILL DRAFT + FILL SHEET READY)
L-07	Each of the four members turns in the PDF on MS Teams	Each	Four "Turned in" confirmations	HUMAN
L-08	Open the PR of our group folder into the course monorepo (`github.com/sunny-nanade/NMIMS-MPSTME-MDRIIA-2026`), touching only `Group_05_…/`	E043	PR open, checks green, no MP4/venv	PREPARED; RUN AFTER MERGES
L-09	Fill §1.1 claim table: verify or Correct/Delete "72 % latency", "11-month CapEx recovery", "90-s verification"; also drop the README/roster "45 → 8.5 min" unless measured	E077 + E043	Each row has source type + verdict	DRAFTED (CLAIMS TABLE)
L-10	Declare all AI use (§5.1 three cases, §6 AI-drafted share per member)	All	Honest entries signed in sign-off	DRAFTED; EACH MEMBER SIGNS

P1 — needed for full marks in a criterion

ID	TASK	OWNER	DONE WHEN	STATUS 30 OCT
L-11	Set timestep 0.002 s; re-check hover, patrol, separation test; write friction/solref justification	E043	Tests pass at 0.002 s	DONE
L-12	500 Hz per-run telemetry with reference, tracking error, actuator forces, FSM state	E043	Row interval = 0.002 s	DONE
L-13	Per-run tracking RMSE and $\int \tau^2 dt$ in the trials CSV	E043	Two new columns	DONE
L-14	Implement `detect_intruder_hsv` + `inverse_pinhole_projection`; measure IoU and latency, noon vs twilight	E051	`vision_illumination_benchmark.csv` written	DRAFTED (E051 PATCH)
L-15	Implement `evaluate_geofence_status` (ray casting, 4 tiers, hysteresis, noise) + breach log	E075	Unit cases for tiers 0–3 pass	DRAFTED (E075 PATCH)
L-16	Wire the surveillance FSM; log state per row; overlay in viewer	E043	One live DETECT→ALERT transition	DONE
L-17	`analytics/statistical_validation.py`: Welch t, df, p, Cohen's d, 95 % CI	E075	Numbers match dossier §4 table	DRAFTED (E075 PATCH)
L-18	Regenerate the three figures from real data; replace starter PNGs	E043	No watermark	DONE

ID	TASK	OWNER	DONE WHEN	STATUS 30 OCT
L–19	Business: H_reclaimed, P_payback/T_payback, NCRI/κ, M/M/c (λ, μ, c, ρ), sensitivity ±30 % + break-even, queue graph	**E077**	Two graphs + §3.1 table reproducible	DRAFTED (E077 PATCH)
L–20	FMEA ≥ 4 modes, RPN pre/post, each mitigation shown in sim/code	**E077 + all**	Table in §3.2	DRAFTED (E077 PATCH)
L–21	Record and post the 3–4 min LinkedIn video per Showcase Guide §7; add URL to dossier §5 and `TEAM_ROSTER.json`	**All · E043 edits**	Live URL	HUMAN (SCRIPT DRAFTED)
L–22	Peer-review Group 06 and Group 11 videos	**All**	§5.2 filled	HUMAN (TEMPLATE DRAFTED)
L–23	Standards decisions §1.2 (ISO 10218, ISO 3691-4, ANSI/RIA R15.06, IEC 60601-1, India Drone Rules 2021) + ISO 21384 / ASTM F3322 from the Standards Manual	**E075 + E043**	Yes/No + clause or reason for each	DRAFTED (FILL SHEET)
L–24	Baseline friction §1.3: today's guard patrol metric with a primary source	**E077**	Value + unit + citation	DRAFTED (SOURCED INPUTS)

P2 — quality and viva safety

ID	TASK	OWNER	DONE WHEN	STATUS 30 OCT
L–25	Fresh-clone rehearsal on a second laptop; fix anything missing	**E043**	Viewer runs from scratch	DONE (FRESH CLONE PASSED)
L–26	Reconcile SAP IDs: dossier template lists 70362400058 / …040 / 70362300055 / 70362300043; `TEAM_ROSTER.json` and README list …030 / …067 / …003 / …080	**All**	One set, confirmed with the official list	HUMAN (NO SAP ID TOUCHED)
L–27	E077: fix git e-mail, re-author commits, move economics to `feat/e077-csbs-campus-security`	**E077**	Commits show as Subhankar-Uro	HUMAN (GIT_FIX.MD)
L–28	Assigned branch names for all four members	**All**	Branches exist with ≥ 4 commits each	E043 DONE; 3 MEMBERS HUMAN
L–29	Update PRD/Implementation Plan to the interior scene (v2.1) — optional	**E043**	—	SKIPPED (OPTIONAL)
L–30	Each member rehearses the five standard viva questions (§14)	**All**	30-second whiteboard rule met	HUMAN (VIVA PREP WRITTEN)

Dependency chain

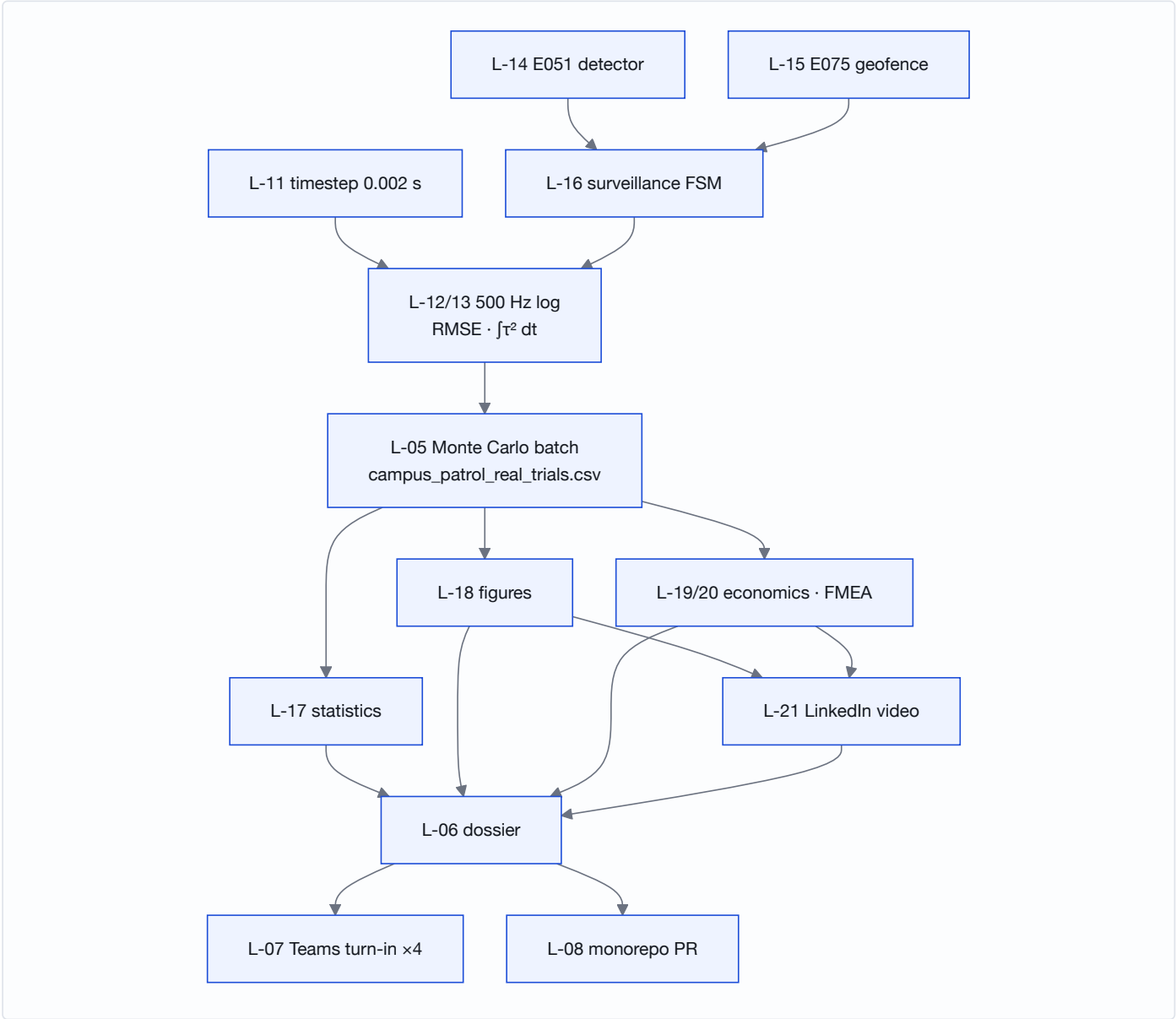


Figure 1 — Critical path. The Monte Carlo batch is the bottleneck: statistics, figures, economics and the video all consume it. If E051/E075 slip, run the batch with detection disabled and report the cycle-time comparison first (see §9 fallback).

7 Work Packages per Member

E043 Muaaz — Lead, flight dynamics, integration, owner of `main`

M–1	Timestep 0.002 s; re-tune if needed; justify friction 1.0/0.01/0.005 and solref 0.004 1 (§2.1)
M–2	500 Hz telemetry columns; per-run RMSE vs waypoint path; $\int \tau^2 \, dt$ over the eight rotor actuators
M–3	Surveillance FSM integration + viewer overlay; merge E051/E075 regions after the ownership gate
M–4	Runner → `campus_patrol_real_trials.csv` ; baseline and proposed conditions; N per §15 ruling
M–5	Repo format (L-02, L-03), monorepo PR (L-08), fresh-clone test (L-25)
M–6	Dossier §2: settings table, gains with units, FSM list, authorship table, ≤ 25-line snippet, FBD worked example (hover: m·g = 1.35 × 9.81 = 13.24 N ⇒ 3.31 N per rotor; max 4 × 7.5 = 30 N ⇒ thrust-to-weight 2.27)
M–7	Video edit and post; compile and export the dossier

E051 Aditya — Computer vision (OpenCV), detection and tracking

A–1	`detect_intruder_hsv` : HSV threshold → morphology → contours → largest bbox; return detected, bbox, latency (ms)
A–2	`inverse_pinhole_projection` : (u, v) + camera pose + depth → world point; report error vs true intruder position
A–3	IoU vs MuJoCo segmentation mask (target ≥ 0.70) and latency at noon vs twilight lighting → `analytics/vision_illumination_benchmark.csv`
A–4	FMEA row “missed intruder at twilight” with measured miss rate; AI-audit case from own work
A–5	≥ 4 commits on `feat/e051-computer-vision-open` , own account

E075 Zaid — Geofencing, telemetry QA (owner of dossier §4)

Z–1	`evaluate_geofence_status` : ray-casting point-in-polygon on the VIP Lounge zone; tiers 0–3 with hysteresis
Z–2	GPS/IMU noise injection (stated σ); JSON breach log with timestamps
Z–3	Optional handled failure for the demo: return-to-home after ≥ 2.5 s telemetry loss (ISO 21384 / ASTM F3322 per Standards Manual §2.4)
Z–4	QA of the batch: logging rate, seeds, N; `statistical_validation.py` ; fill §4 table
Z–5	≥ 4 commits on `feat/e075-restricted-zone-geo-` , own account

E077 Soumya — CSBS business and techno-economics (owner of dossier §3)

S–1	Rewrite `campus_security_economics.py` (see §10); delete constants 48.72, 7.84, 72.0, 50.4, 0.66, 0.31, 35.8, 1.2
S–2	Sourced guard-patrol inputs (walking speed, shift pattern, rounds per shift) or labelled assumptions
S–3	Sensitivity and queue graphs (PNG, 300 DPI) for the dossier and the video
S–4	FMEA table; §1.1 claim verification; §1.3 baseline friction
S–5	Fix commit e-mail; work on `feat/e077-csbs-campus-security` ; park the runtime branch

8 Technical Dossier — Field-by-Field Fill Plan

Every field in the official template, who fills it, and where the value comes from. Fill in Word desktop; never delete a "Select ▾" or "type here" control.

SECTION · FIELD	OWNER	SOURCE OF THE VALUE	STATUS
Cover · roster check	All	Resolve SAP ID mismatch (L-26)	DECISION
§1.1 Claim verification (3 rows)	E077, E043	Primary sources or Correct/Delete; simulation measurement	DRAFTED (FILL SHEET)
§1.2 Standards (5 rows)	E075, E043	Faculty hints: four "No", India Drone Rules "check"; add ISO 21384 / ASTM F3322	DRAFTED (FILL SHEET)
§1.3 Baseline friction	E077	Published guard-patrol metric with citation	DRAFTED (FILL SHEET)
§2.1 Timestep, friction, solref, law, gains, FSM, loads?	E043	Final MJCF + controller constants	PARTIAL
§2.2 Authorship table	All	Git history; scaffold files = Modified, generators = Written	DRAFTED (FILL SHEET)
§2.3 Code snippet ≤ 25 lines (text, Consolas 8 pt)	E043	FSM transition or attitude law with line numbers	DRAFTED (FILL SHEET)
§2.4 FBD + worked equation	E043	Hover-thrust example (§7 M-6), telemetry column reference	DRAFTED (FILL SHEET)
§3.1 Business table, queue model, λ μ c, T_payback, reproducible?	E077	Rewritten script	DRAFTED (FILL SHEET)
§3.2 FMEA (2 seeded + ≥ 2 more) + two graphs	E077 + all	Separation test, vision benchmark, geofence log	DRAFTED (FILL SHEET)
§4 Protocol: baseline, N, distributions, Hz, seeds, CSV path	E075	Runner provenance JSON	DRAFTED (FILL SHEET)
§4 Results table: RMSE, $\int \tau^2\,dt$, primary metric, test, t/df/p, d, CI	E075	`statistical_validation.py`	DRAFTED (FILL SHEET)
§4 Trajectory plot + box-plot	E043	Regenerated figures	DRAFTED (FILL SHEET)
§4 Data Provenance Declaration (4 boxes)	All	Tick only what is true	DRAFTED (FILL SHEET)
§5 Video URL, length, three yes/no checks	E043	LinkedIn post	HUMAN
§5.1 AI audit AI-01…AI-03	All	Real cases caught by docs/paper/running code	DRAFTED (FILL SHEET)
§5.2 Peer review: Group 06 (Crawler Solar Cleaner), Group 11 (Solar PV Inspection AMR)	All	Their LinkedIn videos	HUMAN
§6 Learning record (hours, role match, commit share, AI share, …)	Each member own row	`git shortlog -sn -- Group_05_*/`	HUMAN
Sign-off (original, code runs, viva-ready, signature, consent)	Each member	Personal	HUMAN

TEMPLATE FACTS TO CORRECT IN §2. The pre-filled text says "9 actuators (2 × 4 rotors + 1)" and "PATROL (Alpha NE / Bravo SW sectors)". Our model has **8** rotor actuators and the interior patrol is Alpha on the ground-floor loop and Bravo on the atrium galleries. State the real values in the "type here" fields.

9 Experimental Design (as run)

ITEM	SPECIFICATION
Scenes	Both scenes everywhere: `--scene interior` (inside the Main Building) and `--scene exterior` (the outdoor campus compound inside the boundary wall). All patrolling is inside the college.
Conditions	`guard_baseline` (modelled foot patrol of the same route, speed and stops from cited sources), `uav_sectors` and `uav_offset_ring` (dual UAV, surveillance FSM available)
N and timestep	N = 30 per cell in 6 cells (180 runs) at timestep 0.002 s (500 Hz); N = 50 was not run because the batch budget did not allow it
Randomised inputs	Mean wind and gust sigma, intruder appearance time, dock jitter; one seed per run, recorded in `campus_patrol_real_trials.csv` and its provenance JSON
Primary metric	`worst_revisit_ground_s`, computed by one shared function for all conditions
Other measures	Detection latency and verify time (only when detected), minimum inter-UAV distance, tracking RMSE, control effort, energy, obstacle clearance and contacts
Logging	500 Hz per-run telemetry (gitignored) plus one summary row per run
Statistics	Welch two-sided t-test, df, exact p, Cohen's d, 95 % CI: drafted in E075's `analytics/statistical_validation.py` (patch); every number is computed from the CSV
Detection columns	Detection runs need E051's detector. On the E043 branch alone the detector stub raises NotImplementedError, the FSM stays in PATROL and those columns stay empty ("not measured")

10 CSBS Business Model Still to Build

All quantities dimensionless or in hours; no monetary values anywhere. Every input is either sourced (author, year, DOI/URL) or labelled *assumption* and swept in the sensitivity graph.

QUANTITY	DEFINITION (AS IN THE DOSSIER)	INPUTS FROM
H_reclaimed	Guard-hours per day moved from walking rounds to incident response	Shift pattern (sourced/assumed) × fraction of rounds replaced (from measured cycle times)
P_payback	CapEx / (T_life · ΔOpEx_annual); P < 1 ⇒ pays back within life	CapEx parity factor, T_life, ΔOpEx — all as ratios to the guard baseline
T_payback	CapEx / ΔOpEx, in months	Same
NCRI, κ	NCRI = (C_baseline − C_robot) / C_baseline; κ = 1 − NCRI (manual target κ ≤ 0.25)	Normalised cost components
Queue	M/M/c: λ incidents/h, μ verifications/h per responder, c responders; ρ = λ/(cμ) < 1; W_q via Erlang C	λ assumption; μ from simulated verification time
ΔFTE (manual)	ΔFTE = annual reclaimed hours / 2080	H_reclaimed × 365
Sensitivity	Most uncertain input ±30 %; plot P_payback and NCRI; mark break-even	Script sweep

FMEA seed list (≥ 4 required)

FAILURE MODE	MITIGATION ALREADY IN, OR TO ADD	EVIDENCE
UAV–UAV conflict at crossing (d < 2.5 m) — seeded	Altitude layers + closing-speed governor + sidestep	Separation test 0.44 m → 3.31 m
Missed intruder at twilight (low IoU) — seeded	HSV range / exposure tuning (E051)	Vision illumination benchmark
Wind gust drives UAV into wall	Tilt limiter, velocity PI, corridor routing	Collision counter = 0 across trials
Telemetry / link loss	Return-to-home after ≥ 2.5 s (E075, optional)	Demo run
False geofence alarm from GPS noise	Hysteresis on tier changes (E075)	Breach log with noise on/off

11 Repository & Git Compliance

REQUIRED BY SIR	OUR REPO NOW	ACTION	STATUS
`analytics/campus_patrol_real_trials.csv`	Present: 180 rows, N = 30 in 6 cells, timestep 0.002 s, provenance JSON	L-03, L-05 done	DONE
`analytics/SIMULATED_campus_patrol_benchmark.csv`	Renamed; figure script reads only the real-trials file	L-03 done	DONE
`docs/MDRIIA_Group_05_Technical_Dossier.docx` + `.pdf`	Official template copied; AUTOFILL draft and fill sheet present; filled Word file and PDF are not	L-06 (human, in Word desktop)	HUMAN
`docs/README.md` , `docs/figures/README.md`	Present	L-02 done	DONE
`models/campus_perimeter_patrol.xml` , `src/aerial_patrol_swarm.py` , `src/test_env.py` , `analytics/campus_security_economics.py` , `generate_paper_figures.py`	Present (names preserved)	—	DONE
Extra files (scene generators, viewer, tests, planning)	Present	Allowed inside the group folder; keep MP4s out of the PR	CHECK
Branches `feat/e043-lead-uav-flight-dyna` , `feat/e051-computer-vision-open` , `feat/e075-restricted-zone-geo-` , `feat/e077-csbs-campus-security`	E043 branch has more than 4 commits; the other three branches do not exist yet (patches are in `_handoff/` , not committed)	L-28: each member creates their branch and applies their patch	HUMAN
Conventional commits, ≥ 4 per student, own account	E043: all commits from the noreply author; teammates' commits are theirs to make (`COMMITS.md` in each packet)	L-27, L-28	HUMAN
Work submitted to the course monorepo via PR into `main`	Export prepared; PR not opened	L-08 after teammates' branches are merged and the dossier PDF exists (`docs/submission/monorepo_pr_steps.md`)	HUMAN
No direct commits to `main` (Standards Manual §5)	Initial commit made directly to our private `main`	All further work via PRs	NOTE

12 Integrity Risks & Blockers

#	RISK	IMPACT	RESPONSE
R1	Dossier and video deadlines passed	Possible late penalty or no marks	Send the drafted message (L-01); Q1 is still unanswered; submit as early as possible
R2	Scaffold `np.random` numbers or the README's "45 → 8.5 min" reused as results	Zero for the component (misconduct)	Figure gate; rename to `SIMULATED_` ; quote only real-trials numbers
R3	Hard-coded economics constants	C2 zero	L-04
R4	Monetary units in E077's work	Zero-Money Rule breach	Ratios only
R5	Undeclared AI use	Integrity breach	Full §5.1 / §6 disclosure
R6	Attribution mismatch: work in git ≠ declared role; commits from wrong e-mail	Viva marks; integrity question	Each member commits own region from own account; honest §2.2 and §6
R7	E051/E075 regions still empty two days before viva	C1 control + two members' viva	Handoff patches exist and pass their gates; members apply, commit and be ready to explain them (viva prep files)
R8	Batch runtime too long for N ≥ 30–50 at 900 s each	No statistics	Resolved: N = 30 per cell, 180 runs, dt 0.002 s; N = 50 not run
R9	Indoor scene vs "perimeter patrol" title	Problem-framing / mechanism-match marks	Resolved by faculty ruling: both scenes, all patrolling inside the college (§15)
R10	YOLO (AGPL-3.0, PyTorch) instead of OpenCV	Changes the research question	Resolved by faculty ruling: OpenCV HSV primary, YOLO optional inside E051's draft only
R11	Disk space on the lead laptop is nearly full (≈ 0.3–0.5 GB free)	Batch runs / video export may fail	Free space before running the batch and recording

13 Plan to Saturday 31 October

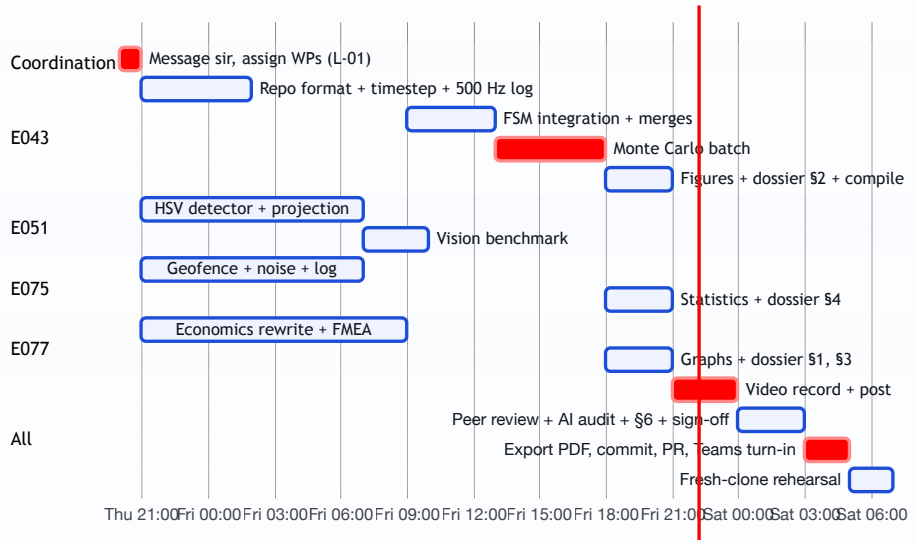


Figure 2 — Compressed schedule. Start times are targets; the batch (M-4) gates the statistics, figures, economics graphs and video. If it slips, record the video with the cycle-time results only.

14 Viva & Expo Readiness

QUESTION SIR WILL ASK (SHOWCASE GUIDE §11)	WHO ANSWERS	PREPARED ANSWER SOURCE
Why that solref time constant and friction? What happens if you halve them?	E043	§2.1 justification; try it live in the viewer
Show one FSM transition live and the line of code that triggers it	E043 (+ E051 DETECT, E075 ALERT)	Viewer overlay + §2.3 snippet
Which input moves payback most, and where does its value come from?	E077	Sensitivity graph + source
Why Welch's t-test? What does Cohen's d mean in plain words?	E075	Unequal variances; d = gap in pooled-SD units
What did AI get wrong, and how did you prove it?	Each member	§5.1 cases

Expo checklist

- [ ] Laptop with simulation running from a fresh `git clone` + `pip install -r requirements.txt`
- [ ] FSM states visible in the viewer; one handled failure (head-on UAV conflict resolved by the governor and sidestep)
- [ ] Business sensitivity and queue graphs open
- [ ] Webcam picture-in-picture of every member; no PPT slides
- [ ] Each member can explain their code in 30 seconds on a whiteboard

15 Faculty Rulings and Open Questions

Rulings given on 29 October 2026 replace the proposals in PRD v3.0. They override earlier text in this document and in PRD v2.0.

#	QUESTION	RULING / STATUS
Q1	Will a late dossier (after 26 Oct) be accepted, and by when?	**OPEN** Unanswered. The drafted message is `docs/submission/email_to_sir_DRAFT.md`; the lead sends it.
Q2	Indoor patrol vs the "campus perimeter patrol" title	Both scenes are required everywhere (`--scene interior` and `--scene exterior`). Interior drones must sense all obstacles (FR-18). All patrolling is inside the college: interior is inside the Main Building, exterior is the outdoor compound inside the boundary wall.
Q3	N ≥ 30 or N ≥ 50?	N = 50 per cell if the budget allows, otherwise 30. We ran N = 30 per cell.
Q4	Timestep 0.002 s or 0.004 s?	0.002 s is mandatory. All models and runs use it.
Q5	OpenCV or YOLO?	OpenCV HSV is the primary detector. YOLO is optional and lives only inside E051's draft.
Q6	Correct SAP IDs	Each member enters their own SAP ID. No SAP ID was written, picked or edited in the repository.

Prepared by E043 for Group 05. v3.1 status reflects the E043 branch `feat/e043-lead-uav-flight-dyna` as of 30 Oct 2026, the three handoff packets in `_handoff/` (not committed), and the audits of that date. v3.0 was based on commit `c3fdea7` (main) and the course monorepo as read on 29 Oct 2026.